

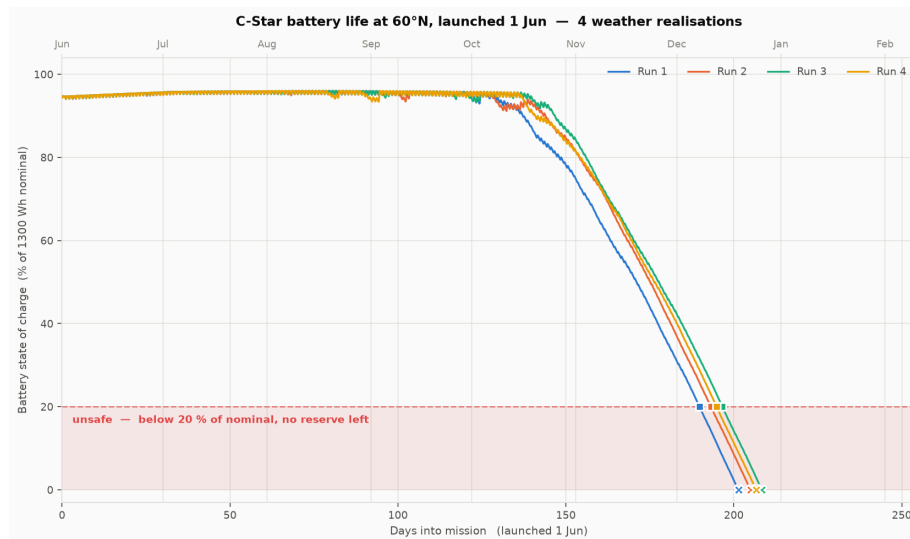
Alternative Power for C-Star at High Latitude

1. Sizing the problem

A customer wants to operate C-Stars at high latitude, where solar power may be limited. Before comparing harvesters I modelled the baseline vehicle, because "would this help?" has no answer until you know how large the deficit is and *when* it falls.

The model tracks battery state of charge hour by hour over a year. What it predicts depends mainly on **latitude, time of year, cloud, wind and sea temperature**: sun angle and day length set how much light reaches a flat deck panel, cloud and sail shading remove more, and cold water quietly reduces usable battery capacity. Cloud and wind are generated as *persistent* random series, so bad spells last days rather than averaging out — which is what actually threatens a vehicle. This is a sizing tool, not a truth machine: it is only as good as its assumptions (§4), it should be improved before any hardware commitment, and **it is no substitute for real sea trials**.

Safe mission duration here means time from launch until the battery first falls below 20 % of nominal — not failure, but the point at which no reserve remains for a storm or a bad weather run. All runs assume a **1 June launch**: it exercises a full seasonal cycle and is the most favourable date, so every failure shown is a genuine capability limit rather than bad timing.



Battery state of charge at 60°N, launched 1 June — four weather realisations

The figure shows four runs at 60 °N. Through summer the battery sits near full and surplus is simply dumped; from late September it falls away, and the vehicle becomes unsafe after **190–197 days**. The runs differ only slightly because at this latitude the answer is set by sun geometry, not weather luck — predictable, but with no good year to hope for.

The problem is darkness, not energy. At 65 °N the array generates 15.0 kWh a year against an 8.8 kWh load — a 1.7× surplus — and the vehicle still dies, because the energy arrives in the wrong months. December generation is 0.1 Wh/day against 24 Wh/day consumed, roughly one-thousandth of July. Day length collapses, a low sun striking a flat deck loses most of its energy to the cosine of the incidence angle, and the beam crosses far more atmosphere. This is a **seasonal storage problem**, so a useful solution must deliver *in the dark months specifically*.

Sweeping latitude, every run survives the full year up to **48 °N**; 50–52 °N is a knife edge where some weather years survive and some do not; **from 54 °N upward every run fails**. The sweep stops at 74 °N: beyond ~75 °N the Arctic is ice-covered, a 1 m sailing hull cannot operate in ice, and the model has no ice physics. Ice also damps waves, which matters below.

The requirement. Solving for the constant extra generation that removes every blackout gives **0.25 W at 65 °N and 0.48 W at 70 °N** — 25–50 % of the vehicle load. If C-Stars are to run **propeller-driven** at any point in a mission, that draw is far larger than the 1 W housekeeping load, so we should aim to **exceed** this figure with real margin rather than just meet it.

The goal. More endurance is always useful, but the ideal is that a C-Star **survives the whole winter**: reaching spring with charge in hand returns it to surplus conditions, so it can keep going — bounded then by battery calendar life, biofouling and equipment failure rather than by the energy balance. Options that fall short still have value; options that clear the winter change what missions are possible.

2. The options

From the brief: a small wind turbine and a wave-energy harvester. **Added for completeness:** a larger battery and propeller regeneration. I added these because any recommendation to spend money on new hardware has to beat the cheap alternatives first — more cells need no new technology at all, and the propeller is already fitted, so both are the

obvious objections a reviewer would raise.

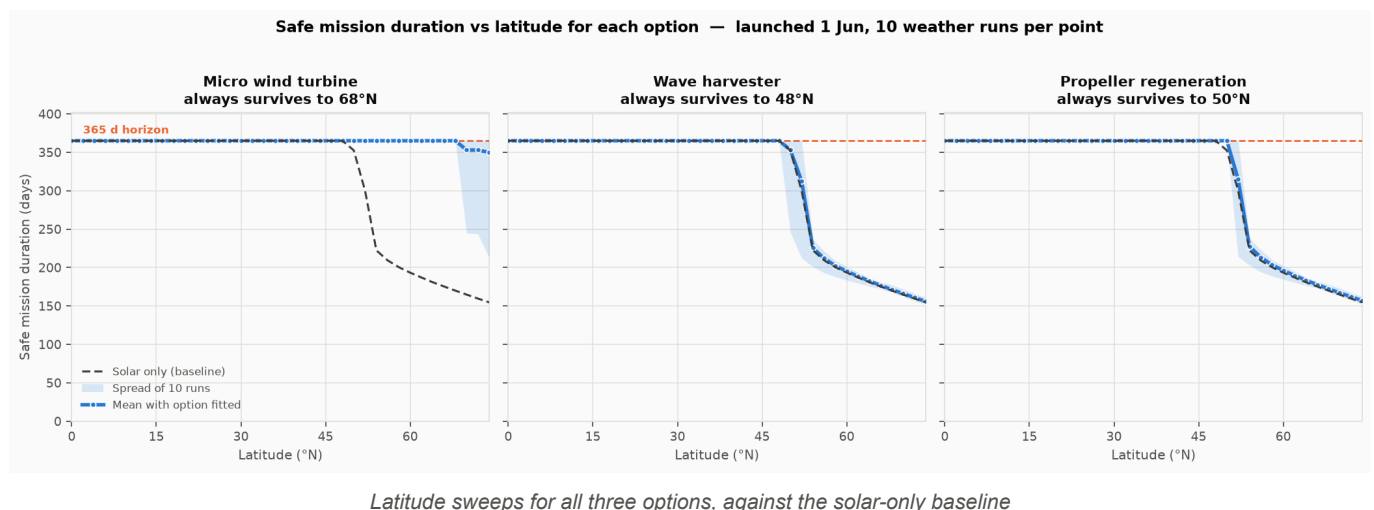
A larger battery — fails on mass. *(added)* Capacity buys endurance linearly at 3.9 days per kg, so there is no efficiency argument against it. But the pack needed to clear winter scales badly: **3640 Wh at 60 °N (+21 kg, 53 % of vehicle mass), 4680 Wh at 65 °N (+31 kg, 77 %), 6500 Wh at 70 °N (+47 kg, 118 %)**. At 70 °N the cells alone outweigh the complete vehicle — not a modification but a different boat. It is a reasonable answer for the marginal 50–52 °N band and nothing beyond, and it adds no generation: it enlarges the tank without addressing what fills it in December.

A wave-energy harvester — falls short by ~20×. *(from the brief)* A 1 m hull is far shorter than an ocean wavelength (50–150 m), so it is a *wave follower* — it rides the surface, leaving no hull-to-water relative motion to exploit. Power can then only come from an internal proof mass reacting against the hull's own motion, and is proportional to the stroke available, of which a 1 m hull has very little. Modelled with a 2 kg mass and 60 mm of stroke, mean output is **0.023 W**, about 2 % of the load. Reaching 0.5 W would need roughly a 5 kg mass moving 0.5 m inside a 1 m vehicle. **This is a scale problem, not a maturity problem.**

More imaginative geometries do escape the wave-follower trap by reaching outside the hull — a **tethered submerged body** (a drogue, plate or small buoy) held deep enough to sit in slower water, so the hull moves relative to it and the tether does work; or a towed flexible membrane. These are genuinely promising in principle and are how the concept would have to be done. I have not assessed them here because they add a deployed external line, a winch or damper, and a large snag, chafe and entanglement risk on an unattended year-long mission — high complexity for a platform whose main virtue is that nothing hangs off it. **With a longer programme they arguably deserve assessment**, and I would not close the door on wave energy purely on the internal-mass result.

Propeller regeneration — self-limiting. *(added)* Let the existing propeller freewheel and run the motor as a generator. Water is 830× denser than air, so a small disc looks promising. But Oshen's propeller has ~2–3 cm blades, and at a 1.5 kn sailing speed mean output is **0.044 W**, far below requirement, while the turbine still costs **14 % of boat speed** (1.50 → 1.29 kn). It cannot be scaled out of: the energy comes from the boat's motion, so a larger disc slows the boat, and power falls with the cube of speed. Output saturates around 0.17 W while the speed penalty grows without limit. Worth enabling opportunistically since it is nearly free, but not a solution.

A micro wind turbine — works, with margin. *(from the brief)* Ocean wind rises with latitude and **peaks in midwinter**, in direct antiphase with the sun — the physical basis of the concept. A 0.20 m rotor with a 3 m/s cut-in, furling in survival winds, gives **2.6 W at 60 °N and 3.1 W at 70 °N** — five to ten times requirement and 2–3× the whole vehicle load, with genuine headroom for propeller-driven operation. Output is reduced by the **atmospheric boundary layer**: wind is quoted at 10 m, but a hub about 1 m above a sea surface that is slowing the flow by friction sees roughly 22 % less.



Compared side by side, only the wind turbine moves the curve: it survives to **68 °N** against 48 °N for solar alone. The wave and propeller traces sit almost exactly on the baseline.

3. Recommendation, integration and testing

Recommend the wind turbine. It is the only option meeting the requirement with margin, the only one working beyond ~50 °N, and the only one leaving enough headroom to use propeller drive.

Integration is the hard part, not the power. A rotating machine competes for masthead space with a *rotating wingsail* — a geometric and aerodynamic clash; mass and windage go high up, attacking righting moment on a 40 kg hull; it adds the vehicle's first exposed bearing, in salt spray, unattended for a year; it must survive knockdown and immersion; and **icing, at**

exactly the latitudes where it is most needed, is the largest unquantified risk. My starting concept is a stern-mounted pylon rather than masthead, accepting lower wind speed to stay clear of the sail and keep the CG down, feeding the existing MPPT input through a rectifier.

De-risking, cheapest and most decisive first. Each test is designed to *kill* the concept early:

1. **Fix the input data (days, ~£0).** Replace the synthetic cloud and wind climatology with ERA5 / NASA POWER reanalysis for the customer's real operating box and re-run. Every number here moves.
2. **Instrument a hull already going to sea (weeks, low cost).** IMU, GPS speed and panel current — retiring three assumptions at once: the acceleration spectrum settles wave, the speed distribution settles regeneration, panel current gives real shading loss.
3. **Wind-tunnel or CFD the sail-rotor interaction (~3 weeks).** The question that decides whether the concept is integrable at all.
4. **Bench-test a candidate rotor (~2 weeks), then cold-soak and ice test it** — the risk most likely to be found too late.
5. **Sea trial as a data logger, not a mission (~1 month).** One vehicle, everything logged, on a recoverable route. No customer mission until a full seasonal dataset exists.

Two months vs six. With **two months** the goal is a *decision*, not a prototype: steps 1–4 plus a deliberately non-flight-representative lash-up — an off-the-shelf marine turbine on a stern pylon, logged in sheltered water. The deliverable is a measured power curve and a defensible answer on sail interaction. With **six months** the first two are unchanged — those tests are on the critical path either way. The extra four buy a sealed, bearing-life-tested installation, the icing and knockdown survival campaign, a winter deployment above 60 °N validating the model against reality, and re-qualification of stability with mass aloft — plus enough room to assess a tethered wave harvester properly. The distinction matters: **the two-month plan is not a compressed six-month plan.** It drops qualification and buys a decision rather than a product.

Next step. Two weeks of reanalysis data and one instrumented deployment — a few hundred pounds — would firm up every number here before any hardware commitment.

4. Assumptions and limitations

The model is a screening tool. Its main limitations: cloud and wind come from latitude correlations rather than measured data; there is no sea-ice, storm-survival or biofouling model; the vehicle is assumed to hold a fixed load with no adaptive power management; wave power is assessed only for an internal proof mass; and drag, stability and seakeeping effects of any added hardware are outside it. It predicts *relative* differences between options far better than absolute endurance, and **no version of it replaces a winter at sea.**

#	Assumption	Why it matters	To firm up
A9	Cloud from a latitude correlation	Sets the entire solar deficit	ERA5 / PVGIS for the real box
A17	Wind climatology, winter peak	The whole case for wind	ERA5 reanalysis
A21	Wave stroke and proof mass	Decides a 20× shortfall	IMU on a hull at sea
A23	1.5 kn sailing speed	Regen power scales as speed cubed	GPS traces from real missions
A24	~5 cm propeller disc; output peaks near 0.17 W at ~10 cm before drag losses overtake it	Bounds regeneration at any size	Tow tank: power and drag together
A2	Panels flat on deck	Low-sun cosine loss is punishing	CAD; check for usable vertical area

Two further assumptions bound the study and should be confirmed with Oshen: that **no additional solar area is available**, and that **average consumption cannot be reduced below 1 W**. Since the deficit is measured in tenths of a watt, both are disproportionately powerful levers — a winter duty-cycle reduction may well be cheaper than any hardware considered here.

Supporting work: `cstar_power_model.py`, `run_mission.py`, `sweep_latitude.py`, `sweep_battery.py`, `sweep_compare.py`.